MTH636 – Game Theory

Lecture 6: Two-Player Zero-Sum Games

1 Zero-Sum Games: Max-Min and Min-Max

Recall that in a two-player zero-sum game, $u_1(s_1, s_2) + u_2(s_1, s_2) = 0$. Let $u = u_1$. Thus, $u_2 = -u$. Player 1 wants to maximize u, while Player 2 wants to maximize $u_2 = -u$, which is equivalent to Player 2 wanting to *minimize* u.

Example: Zero-Sum Matrix Notation

Consider a game where Player 1 chooses U or NU, and Player 2 chooses R or NR. Let the payoffs for Player 1 be: $u(U, R) = 3$, $u(NU, R) = -2$, $u(U, NR) = 1$, $u(NU, NR) = 3$.
We represent this simply with Player 1's payoffs:

$P_1 \setminus P_2$	R	NR
U	$(3, -3)$	$(1, -1)$
NU	$(-2, 2)$	$(3, -3)$

1.1 Player 1: Max-Min Value ($\underline{v}$)

Player 1 guarantees themselves a payoff of:

$$\underline{v}_1 = \max_{s_1 \in S_1} \min_{s_2 \in S_2} u_1(s_1, s_2) = \max_{s_1} \min_{s_2} u(s_1, s_2)$$

We denote this as $\underline{v}$. This is the **maximum amount Player 1 can guarantee to get**.

1.2 Player 2: Min-Max Value ($\overline{v}$)

Player 2 looks at their own max-min value:

$$\underline{v}_2 = \max_{s_2 \in S_2} \min_{s_1 \in S_1} u_2(s_1, s_2) = \max_{s_2} \min_{s_1} (-u(s_1, s_2))$$

Because $\max(-f) = -\min(f)$, we can rewrite this as:

$$\underline{v}_2 = -\min_{s_2 \in S_2} \max_{s_1 \in S_1} u(s_1, s_2)$$

Let $\overline{v} = \min_{s_2} \max_{s_1} u(s_1, s_2)$. This is called the **min-max value**. For Player 2, $\overline{v}$ represents the **minimum amount Player 2 can guarantee to pay** to Player 1. In this context, for Player 2, we call the max-min strategy the **min-max strategy**.

2 The Relationship Between $\underline{v}$ and $\overline{v}$

Theorem: Inequality of Max-Min and Min-Max

In general, for any game:

$$\underline{v} \leq \overline{v}$$

Meaning: (The max amount P1 can guarantee to get) $\leq$ (The min amount P2 can guarantee to pay).

Example: Example 1: Value Exists

	L	C	R	**Row Min**
T	3	-5	-2	-5
M	1	4	1	**1** $\Longrightarrow \underline{v} = 1$
B	6	-3	-5	-5
Col Max	6	4	**1**	$\Longrightarrow \overline{v} = 1$

Here, $\underline{v} = 1$ and $\overline{v} = 1$. The max-min strategy for P1 is M, and the min-max strategy for P2 is R.

Example: Example 2: No Value in Pure Strategies

	L	R	**Row Min**
T	-2	5	-2
B	3	0	**0** $\Longrightarrow \underline{v} = 0$
Col Max	**3**	5	$\Longrightarrow \overline{v} = 3$

Here, $\underline{v} = 0$ and $\overline{v} = 3$. Since $\underline{v} < \overline{v}$, this game has **no value in pure strategies**.

3 Value of a Game and Optimal Strategies

Definition: Value of a Game

For a two-player zero-sum game, if $\underline{v} = \overline{v} = v$, then we say the game has a **value** v. And, the max-min and min-max strategies are called **optimal strategies**.

Suppose s_1^* is a max-min strategy for P1, and s_2^* is a min-max strategy for P2. We always have:

$$\underline{v} \leq u(s_1^*, s_2^*) \leq \overline{v}$$

If the game has a value ($\underline{v} = \overline{v} = v$), then $u(s_1^*, s_2^*) = v$. Furthermore:

$$u(s_1, s_2^*) \leq v \quad \text{and} \quad u(s_1^*, s_2) \geq v$$

4 Nash Equilibrium Connection

Definition: Nash Equilibrium (NE)

A strategy profile $(s_1^*, \ldots, s_n^*)$ is called a **Nash Equilibrium** if no player has an incentive to unilaterally deviate. For all i:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i \in S_i$$

Theorem: Minimax Theorem (Partial Statement)

For a two-player zero-sum game, if the value exists ($\underline{v} = \overline{v}$), then (s_1^*, s_2^*) is a Nash Equilibrium, where s_1^* is a max-min strategy of Player 1, and s_2^* is a min-max strategy for Player 2.

Key Takeaway

In zero-sum games with a defined value, playing defensively (optimizing your worst-case scenario) coincides perfectly with finding the Nash Equilibrium. Your best defensive move is also your best offensive response to an optimal opponent!